

Session 4 — Analysis II

Solutions

As in Sessions 1–3, these solutions are written to be read on their own. Each exercise opens with a shaded box recalling the one definition or rule it rests on, and why that rule is what it is; then the computation is carried out with every intermediate step visible; then the result is read in words. Many exercises end by naming the mistake the question was built to catch. Session 3 asked one question — how fast is this quantity changing? — for a single number changing in time. Session 4 asks it when there is more than one number: several inputs (partial derivatives, gradient), several outputs (velocity, Jacobian), and finally several quantities whose rates are prescribed in terms of each other (a linear system, and its eigenvalues).

The three objects of this session, and the name of each. Three notations are used throughout, and it is worth fixing the words for them before anything else:

$f'(t)$	the derivative of f	a <i>number</i> : the slope of $f : \mathbb{R} \rightarrow \mathbb{R}$ at t
$\nabla f = (\partial_x f, \partial_y f)$	the gradient of f	a <i>vector</i> , for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$
$\mathbf{J} = \begin{pmatrix} \partial_x F_1 & \partial_y F_1 \\ \partial_x F_2 & \partial_y F_2 \end{pmatrix}$	the Jacobian of $\mathbf{F}$	a <i>matrix</i> , for $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

Read the middle column aloud: f' is “ f prime”, the derivative; ∇f is “nabla f ” or “grad f ”, the gradient; $\mathbf{J}$ is the Jacobian *matrix* of $\mathbf{F}$ (never of a number — if $\mathbf{F}$ has a single output, its Jacobian is one row, and that row is the gradient: see Exercise 10(d)). A dot on top, as in $\dot{x}$, always means d/dt .

The three are the same idea at three sizes. Each is the *best linear approximation* of its object at a point, and each is used in the same way:

$$f(t_0 + h) \approx f(t_0) + f'(t_0)h, \quad f(\mathbf{x}_0 + \boldsymbol{\delta}) \approx f(\mathbf{x}_0) + \nabla f(\mathbf{x}_0) \cdot \boldsymbol{\delta}, \quad \mathbf{F}(\mathbf{x}_0 + \boldsymbol{\delta}) \approx \mathbf{F}(\mathbf{x}_0) + \mathbf{J}(\mathbf{x}_0) \boldsymbol{\delta}.$$

The classification used constantly in Part C. For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ in the plane, with $T = \text{tr } \mathbf{A}$ and $D = \det \mathbf{A}$, the eigenvalues are the roots of $\lambda^2 - T\lambda + D = 0$, i.e.

$$\lambda_{\pm} = \frac{T \pm \sqrt{T^2 - 4D}}{2}, \quad \text{so} \quad \lambda_+ + \lambda_- = T, \quad \lambda_+ \lambda_- = D.$$

$D < 0$	saddle (one eigenvalue of each sign; always unstable)
$D > 0, T^2 \geq 4D$	node — stable if $T < 0$, unstable if $T > 0$
$D > 0, T^2 < 4D$	spiral — stable if $T < 0$, unstable if $T > 0$
$D > 0, T = 0$	centre (closed orbits)

Stability is decided by T , the shape by the discriminant $T^2 - 4D$, and the existence of a saddle by the sign of D alone.

Exercise 1 Compute $\partial f / \partial x$ and $\partial f / \partial y$ for (a) $x^2 + 3xy + y$; (b) $x^2 + 4y^2$; (c) $x^2 y + \sin y$; (d) $e^{-(x^2 + y^2)}$.

What this exercise uses. A **partial derivative** is an ordinary derivative taken with the other variable *frozen*: to compute $\partial f/\partial x$, treat y as a constant and differentiate in x with the one-variable rules of Session 3. Nothing new is being defined — $\partial f/\partial x$ at (x_0, y_0) is exactly the derivative at x_0 of the one-variable function $x \mapsto f(x, y_0)$, the slice of the surface above the line $y = y_0$. The symbol ∂ instead of d is a reminder that something has been held fixed.

Solution.

- (a) Freezing y , the term $3xy$ has constant coefficient $3y$ and the lone y is a constant, so it vanishes; freezing x , the term x^2 vanishes and $3xy$ has constant coefficient $3x$:

$$\frac{\partial f}{\partial x} = 2x + 3y, \quad \frac{\partial f}{\partial y} = 3x + 1.$$

- (b) No mixed term, so each variable is differentiated on its own:

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 8y.$$

- (c) In x^2y the factor y is a constant for ∂_x and the factor x^2 is a constant for ∂_y ; $\sin y$ does not depend on x :

$$\frac{\partial f}{\partial x} = 2xy, \quad \frac{\partial f}{\partial y} = x^2 + \cos y.$$

- (d) Chain rule in each variable, with the inside $u = -(x^2 + y^2)$ giving $\partial u/\partial x = -2x$ and $\partial u/\partial y = -2y$:

$$\frac{\partial f}{\partial x} = -2x e^{-(x^2+y^2)}, \quad \frac{\partial f}{\partial y} = -2y e^{-(x^2+y^2)}.$$

Both vanish only at the origin, and the two are proportional to x and y : the gradient points radially. That is Exercise 6(c).

Common mistake. Differentiating the frozen variable “a little bit” — writing $\partial_x(3xy) = 3y + 3x$, as though the product rule applied. It does not: y is a constant here, and the product rule with a constant factor is just $(cu)' = cu'$. The check is to substitute a number: with $y = 2$ the function is $x^2 + 6x + 2$, whose derivative is $2x + 6 = 2x + 3y$ ✓.

Exercise 2 (a) For $f(x, y) = x^2 + 3xy + y$, evaluate $\partial f/\partial x$ at $(1, 0)$ and at $(1, 2)$. (b) Why are the two different, and what does that say about functions of several variables?

What this exercise uses. A partial derivative is a *function of both variables*, not a number: $\partial f/\partial x$ depends on y as well as on x . Geometrically, it is the slope of the surface in the x direction, measured at a specific point — and the surface may well tilt differently in the x direction depending on how far along y you are standing.

Solution.

- (a) From Exercise 1(a), $\partial f/\partial x = 2x + 3y$. Substituting:

$$\frac{\partial f}{\partial x}(1, 0) = 2(1) + 3(0) = \boxed{2}, \quad \frac{\partial f}{\partial x}(1, 2) = 2(1) + 3(2) = \boxed{8}.$$

Same x , different y , four times the slope.

- (b) Because the two variables **interact**: the coefficient of the interaction is the mixed term $3xy$, and it is what makes the slope in x depend on y . In one line: *the rate of change in one direction is itself a function of the position in the other directions*.

This is the one genuinely new feature of several variables, and it has a name: the mixed partial $\partial^2 f / \partial y \partial x = 3 \neq 0$ is exactly the measure of it. If the mixed partial were zero everywhere — as for $f = x^2 + 4y^2$ in Exercise 1(b), whose $\partial_x f = 2x$ does not mention y — the two variables would not talk to each other, and the function would be a sum of two independent one-variable problems.

Remark. This is why “the effect of x ” is not a well-defined quantity for a function of several variables: it is a family of numbers, one per value of y . Reporting a single one — a single regression coefficient, say — is a statement about a particular slice, or about a model in which the interaction has been assumed away.

Exercise 3 $f(x, y) = x^2 + 4y^2$. (a) Sketch the contours $f = 1$ and $f = 4$. (b) Compute $\nabla f(1, 1)$. (c) The contour through $(1, 1)$ has tangent direction $(-8, 2)$: verify orthogonality. (d) In which direction from $(1, 1)$ does f increase fastest, and at what rate?

What this exercise uses. A **contour** (level set) of f is a curve $\{f = c\}$: all the points at the same height. The **gradient** is the vector of partial derivatives, $\nabla f = (\partial_x f, \partial_y f)$, and it carries two pieces of information at once:

direction: $\frac{\nabla f}{\|\nabla f\|}$ is the direction of steepest increase, **length:** $\|\nabla f\|$ is the rate along it.

It is *perpendicular to the contour through the point*, and the reason is one line of chain rule: if $\mathbf{c}(s)$ runs along a contour then $f(\mathbf{c}(s))$ is constant, so $0 = \frac{d}{ds} f(\mathbf{c}(s)) = \nabla f \cdot \mathbf{c}'(s)$ — the gradient is orthogonal to every tangent to the contour. Along a curve leaving the point in a unit direction $\mathbf{u}$, the rate of change is $\nabla f \cdot \mathbf{u}$, which is largest when $\mathbf{u}$ is parallel to ∇f .

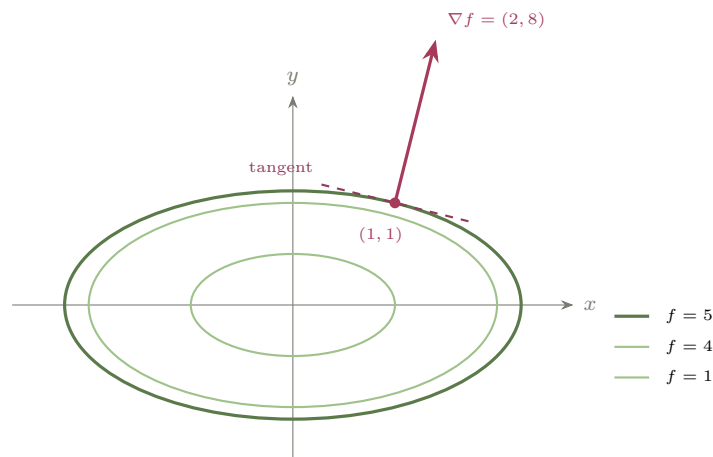
Solution.

- (a) Both contours are **ellipses**, wider than they are tall because the y term is weighted four times more heavily. Writing them in standard form:

$$f = 1 : \quad \frac{x^2}{1} + \frac{y^2}{(1/2)^2} = 1 \quad \text{through } (\pm 1, 0) \text{ and } (0, \pm \tfrac{1}{2}),$$

$$f = 4 : \quad \frac{x^2}{2^2} + \frac{y^2}{1^2} = 1 \quad \text{through } (\pm 2, 0) \text{ and } (0, \pm 1).$$

In general $\{f = c\}$ has semi-axes $\sqrt{c}$ along x and $\sqrt{c}/2$ along y : the two axes grow like $\sqrt{c}$, so doubling the height does not double the ellipse.



Contours of $f = x^2 + 4y^2$. The point $(1, 1)$ sits on the contour $f = 5$; the gradient there is perpendicular to that contour and points *outwards*, towards the higher levels. It is drawn to $\frac{1}{5}$ of its true length.

(b) $\nabla f = (2x, 8y)$, so

$$\nabla f(1, 1) = \boxed{(2, 8)}.$$

(c) The dot product is

$$(2, 8) \cdot (-8, 2) = 2 \times (-8) + 8 \times 2 = -16 + 16 = \boxed{0},$$

so the two vectors are orthogonal ✓. (The tangent direction had to be proportional to $(-8, 2)$, or equally to $(-4, 1)$: rotating $\nabla f = (2, 8)$ by a quarter turn gives $(-8, 2)$.)

(d) Fastest increase is along ∇f itself. Normalising,

$$\mathbf{u} = \frac{(2, 8)}{\|(2, 8)\|} = \frac{(2, 8)}{\sqrt{4 + 64}} = \frac{(1, 4)}{\sqrt{17}} \approx (0.24, 0.97),$$

and the rate along it is the length of the gradient,

$$\|\nabla f(1, 1)\| = \sqrt{68} = 2\sqrt{17} \approx \boxed{8.25}.$$

Reading: moving one unit from $(1, 1)$ in the best direction raises f by about 8.25 — to first order. The direction is much closer to vertical than to horizontal, which is the picture saying that the surface is steeper in y than in x , as the coefficient 4 demands.

Common mistake. Believing that the gradient points “towards the minimum”. It points **uphill**, along the local steepest *increase*; descent methods therefore move along $-\nabla f$. And “towards” is too strong even then: the gradient is a purely local direction, and on a curved surface it does not aim at the bottom, it only gives the best direction for one infinitesimal step.

Exercise 4 $A(d, p) = 1 - e^{-p/(2d)}$, $d > 0$, $p \geq 0$. (a) $\partial A / \partial p$: is accuracy increasing in practice? (b) $\partial A / \partial d$ and its sign, interpreted. (c) Show the contour $A = c$ is a straight line through the origin, and give its slope.

What this exercise uses. Two ideas meet here. First, partial derivatives are read as *signs*: a positive partial means “increasing this input increases the output, the other input held fixed”. Second, an **iso-curve** (a contour) is the set of input combinations giving the same output — here, all the (difficulty, practice) pairs producing the same accuracy. Finding it is solving $A(d, p) = c$ for one variable in terms of the other, and the tool for undoing an exponential is the logarithm: $e^u = v \iff u = \ln v$ (for $v > 0$).

Solution.

- (a) Chain rule with the inside $u = -p/(2d)$, whose p -derivative is $-1/(2d)$ (d frozen):

$$\frac{\partial A}{\partial p} = -e^{-p/(2d)} \times \left(-\frac{1}{2d}\right) = \boxed{\frac{1}{2d} e^{-p/(2d)}}.$$

Both factors are strictly positive ($d > 0$, and an exponential is never zero), so $\partial A/\partial p > 0$: **accuracy increases with practice**, at every level of practice. It increases ever more slowly, though, since the factor $e^{-p/(2d)}$ decays — the familiar diminishing returns.

- (b) Now freeze p and differentiate in d . Write the inside as $u = -\frac{p}{2}d^{-1}$, so that the power rule gives $\partial u/\partial d = \frac{p}{2}d^{-2} = p/(2d^2)$:

$$\frac{\partial A}{\partial d} = -e^{-p/(2d)} \times \frac{p}{2d^2} = \boxed{-\frac{p}{2d^2} e^{-p/(2d)}}.$$

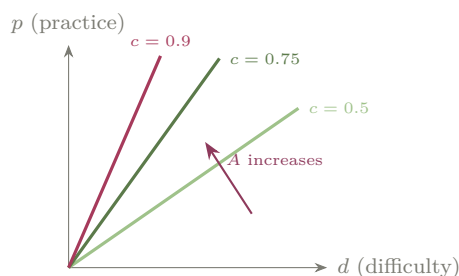
For $p > 0$ this is **strictly negative**; at $p = 0$ it is exactly 0. *Interpretation:* at equal practice, a harder task gives lower accuracy — and its size $\frac{p}{2d^2}e^{-p/(2d)}$ is a growing factor p against a decaying exponential, so it *rises* with practice up to $p = 2d$ and *falls* beyond, tending to 0. Difficulty matters most to a moderately practised participant: with no practice at all ($p = 0$) accuracy is 0 whatever the difficulty — that is the vanishing derivative — and at the other extreme accuracy is near 1 whatever the difficulty. Neither end is an exception to the story; both are consequences of it.

- (c) Set $A = c$ and undo the exponential. With $0 \leq c < 1$:

$$1 - e^{-p/(2d)} = c \iff e^{-p/(2d)} = 1 - c \iff -\frac{p}{2d} = \ln(1 - c),$$

$$\iff \boxed{p = [-2 \ln(1 - c)] d}, \quad \text{slope } \frac{dp}{dd} = -2 \ln(1 - c) = 2 \ln \frac{1}{1 - c}.$$

The right-hand side is a constant times d with no additive term, so the contour is a **straight line through the origin** ✓, of the stated slope. The slope is positive because $0 < 1 - c \leq 1$ makes $\ln(1 - c) \leq 0$; it is 0 for $c = 0$ (the line $p = 0$: zero practice gives zero accuracy at any difficulty) and it tends to $+\infty$ as $c \rightarrow 1$ (perfect accuracy is unreachable in finite practice, so there is no contour $A = 1$ at all).



Iso-performance lines $p = [-2\ln(1 - c)]d$. All pass through the origin; a higher target accuracy is a steeper line, i.e. more practice required per unit of difficulty. Moving up and to the left raises accuracy — the direction of ∇A .

Reading: to hold accuracy fixed, practice must scale *in proportion* to difficulty. Doubling the difficulty of the task doubles the practice needed, whatever the target — only the constant of proportionality depends on how accurate you want to be.

Common mistake. Reporting the sign of $\partial A/\partial d$ without saying what is held fixed. “Harder tasks give lower accuracy” is a statement about *fixed practice*; along an iso-performance line difficulty rises and accuracy does not change at all, because practice rises with it. A partial derivative is meaningless without its *ceteris paribus* clause.

Exercise 5 $J(w) = \sum_{i=1}^n (y_i - wx_i)^2$. (a) dJ/dw . (b) Set it to zero and solve. (c) One step of gradient descent $w \leftarrow w - \eta dJ/dw$; what does η control? (d) Why exactly one critical point, and why a minimum?

What this exercise uses. Fitting a model is minimising an **error surface**: a function of the parameters, whose value is how badly the model reproduces the data. The parameters are the variables; the data x_i, y_i are *constants*. Two ways to find the bottom: solve $\nabla J = \mathbf{0}$ exactly (possible here because J is quadratic), or walk downhill by **gradient descent**, repeatedly taking a small step along $-\nabla J$. The second is what is used when the first has no closed form. Here there is a single parameter, so the gradient is the ordinary derivative dJ/dw .

Solution.

(a) Differentiate the sum term by term; each term is $(\cdot)^2$ with inside $y_i - wx_i$, whose w -derivative is $-x_i$:

$$\frac{dJ}{dw} = \sum_{i=1}^n 2(y_i - wx_i) \times (-x_i) = \boxed{-2 \sum_{i=1}^n x_i (y_i - wx_i)}.$$

The bracket $r_i = y_i - wx_i$ is the i -th **residual**, so the derivative is $-2 \sum_i x_i r_i$: minus twice the *inner product* of the predictor with the residuals.

(b) Setting the derivative to zero and expanding:

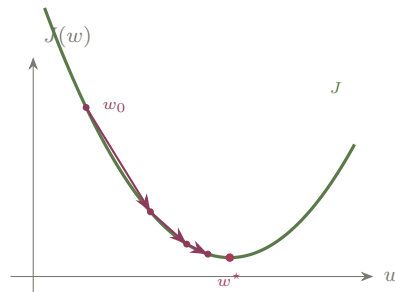
$$\sum_i x_i y_i - w \sum_i x_i^2 = 0 \quad \implies \quad \boxed{w = \frac{\sum_i x_i y_i}{\sum_i x_i^2}},$$

valid whenever $\sum_i x_i^2 \neq 0$, i.e. as long as the predictor is not identically zero. (If it is, every w gives the same error and nothing can be estimated — correctly so.) This is **least-squares regression through the origin**. Note what the condition $dJ/dw = 0$ says in words: at the optimum the residual vector is **orthogonal** to the predictor, $\sum_i x_i r_i = 0$ — everything x could explain has been used. (Orthogonal, not *uncorrelated*: with no intercept in the model the residuals need not have mean zero, and their correlation with x is in general not 0.)

(c) Substituting the derivative into the update rule, the two minus signs cancel:

$$w \leftarrow w - \eta \frac{dJ}{dw} = \boxed{w + 2\eta \sum_i x_i (y_i - wx_i)}.$$

η is the **learning rate**: it fixes how far to move per step, the *direction* being already decided by the sign of the derivative. Too small and the fit crawls; too large and it steps past the minimum and can diverge. This is exactly the step-size trade-off of Euler's method in Session 3 — and it is the same computation: for this quadratic J the update is $w \leftarrow (1 - 2\eta \sum x_i^2)w + \text{const}$, a linear recursion, stable precisely when $|1 - 2\eta \sum x_i^2| < 1$, i.e. $0 < \eta < 1/\sum x_i^2$.



Gradient descent on a one-parameter error surface. The steps shorten as the slope flattens, because the step length is $\eta |dJ/dw|$ — the method slows down on its own approach to the bottom.

(d) Expand J as a polynomial in w , treating the data as constants:

$$J(w) = \underbrace{\left(\sum_i x_i^2\right)}_{>0} w^2 - 2\left(\sum_i x_i y_i\right)w + \sum_i y_i^2.$$

This is a **parabola opening upward**, since the leading coefficient $\sum_i x_i^2$ is a sum of squares and is positive (again assuming not all x_i are zero). A parabola has exactly one stationary point, and for an upward parabola it is the global minimum. Equivalently $d^2 J/dw^2 = 2\sum_i x_i^2 > 0$ everywhere: the error surface is convex, so any critical point is the minimum and there is only one.

Remark. Convexity is what makes least squares comfortable, and it is a property of this particular error function — not of fitting in general. The moment the model is nonlinear in its parameters, the surface acquires several valleys and gradient descent finds the one it happens to start in.

Common mistake. Differentiating with respect to x_i or y_i . The data are given; the unknown is w . A quick way to keep it straight: J is a function of the parameters only, and after fitting you will plot J against w — so w is the axis, hence the variable.

Exercise 6 Find all points with $\nabla f = \mathbf{0}$ and say what kind they are: (a) $f = x^2 + y^2$; (b) $f = x^2 - y^2$ (describe the surface); (c) $f = e^{-(x^2+y^2)}$.

What this exercise uses. A **critical point** is a point where $\nabla f = \mathbf{0}$: the surface is locally flat, its tangent plane horizontal. As in one variable (Session 3, Exercise 7) this is *necessary* for an interior maximum or minimum and not sufficient — and in two variables there is a genuinely new third possibility, the **saddle**: a point that is a minimum along one direction and a maximum along another. The cheap way to classify without any second-derivative machinery is to *restrict f to lines through the point* and compare.

Solution.

(a) $\nabla f = (2x, 2y) = \mathbf{0}$ only at $\boxed{(0,0)}$. There $f = 0$, and everywhere else $f = x^2 + y^2 > 0$: the origin is a **global minimum**. The surface is a bowl (a circular paraboloid), and its contours are circles.

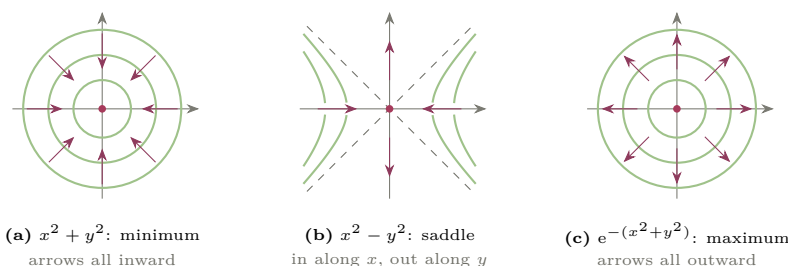
(b) $\nabla f = (2x, -2y) = \mathbf{0}$ only at $\boxed{(0,0)}$, where again $f = 0$. But now the two directions disagree:

along the x -axis ($y = 0$) : $f = x^2 > 0$ (a minimum, uphill both ways),

along the y -axis ($x = 0$) : $f = -y^2 < 0$ (a maximum, downhill both ways).

So the origin is neither: it is a **saddle**. The surface is a *mountain pass* — rising as you walk east or west, falling as you walk north or south. Its contours $x^2 - y^2 = c$ are hyperbolas, and the two lines $y = \pm x$ are the level set $f = 0$ crossing itself at the saddle.

(c) $\nabla f = -2e^{-(x^2+y^2)}(x, y)$, and since the exponential never vanishes this is $\mathbf{0}$ only at $\boxed{(0,0)}$. There $f = 1$; elsewhere the exponent is strictly negative so $f < 1$. The origin is a **global maximum** — a single smooth bump decaying to 0 in every direction, with circular contours.



Contours near each critical point, with blue arrows showing the direction of steepest *descent*, $-\nabla f$ — the way a ball placed there would start to roll, and the direction gradient descent moves in. Inward everywhere: a minimum sits at the centre. Outward everywhere: a maximum. Inward along one axis and outward along the other: a saddle — and that mixture is what has no analogue in one variable.

Remark. (a) and (c) have the same contours — concentric circles — and opposite classifications. A contour plot alone never distinguishes a peak from a pit; you need the values, or the direction of the gradient. This is exactly why contour maps are printed with labelled heights.

Common mistake. Concluding from $\nabla f = \mathbf{0}$ that the point is a maximum or a minimum. Part (b) is the standing refutation, and in two variables the saddle is not an exotic case: it is what generically happens when the two directions disagree. Always test at least two directions before naming the point.

Exercise 7 For each trajectory compute $\dot{\mathbf{x}}(t)$, the speed $\|\dot{\mathbf{x}}(t)\|$, and describe the path: (a) $(2t, 3 - t)$; (b) (t, t^2) ; (c) $(\cos t, \sin t)$; (d) $(e^{-t} \cos t, e^{-t} \sin t)$.

What this exercise uses. A vector-valued function of one variable $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^2$ is differentiated component by component: $\dot{\mathbf{x}} = (\dot{x}_1, \dot{x}_2)$. Two different objects come out of it, and they must not be confused:

velocity $\dot{\mathbf{x}}(t)$ (a vector: direction and rate), **speed** $\|\dot{\mathbf{x}}(t)\| = \sqrt{\dot{x}_1^2 + \dot{x}_2^2}$ (a number).

The velocity is always *tangent* to the path, pointing in the direction of travel. And the path (the set of points visited) is one thing, the motion along it (how fast, in which direction) is another: the same curve can be traced at any speed.

Solution.

(a) $\dot{\mathbf{x}} = \boxed{(2, -1)}$, constant, so $\|\dot{\mathbf{x}}\| = \sqrt{4+1} = \boxed{\sqrt{5} \approx 2.24}$, also constant. The path is a **straight line** through $(0, 3)$ at $t = 0$ with direction $(2, -1)$, travelled at constant velocity. (Eliminating t : $x = 2t$ and $y = 3 - t$ give $y = 3 - x/2$.)

(b) $\dot{\mathbf{x}} = \boxed{(1, 2t)}$ and $\|\dot{\mathbf{x}}\| = \boxed{\sqrt{1+4t^2}}$, which increases with $|t|$. The path is the **parabola** $y = x^2$ (since $x = t$ forces $y = x^2$), traversed left to right: slowest at the vertex $t = 0$, where the speed is 1, and faster the steeper the curve gets on either side.

(c) $\dot{\mathbf{x}} = \boxed{(-\sin t, \cos t)}$ and

$$\|\dot{\mathbf{x}}\| = \sqrt{\sin^2 t + \cos^2 t} = \boxed{1}.$$

The path is the **unit circle**, traversed anticlockwise at unit speed; one full turn takes 2π .

(d) Product rule on each component, with $(e^{-t})' = -e^{-t}$:

$$\dot{\mathbf{x}} = \boxed{(-e^{-t}(\cos t + \sin t), e^{-t}(\cos t - \sin t))}.$$

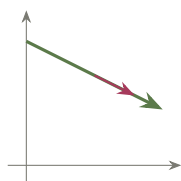
For the speed, expand the two squares — the cross terms $\pm 2 \sin t \cos t$ cancel:

$$(\cos t + \sin t)^2 + (\cos t - \sin t)^2 = 2(\cos^2 t + \sin^2 t) = 2, \quad \text{so} \quad \|\dot{\mathbf{x}}\| = \boxed{\sqrt{2} e^{-t}}.$$

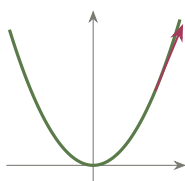
The path is an **inward spiral**: the distance from the origin is

$$\|\mathbf{x}(t)\| = e^{-t} \sqrt{\cos^2 t + \sin^2 t} = e^{-t},$$

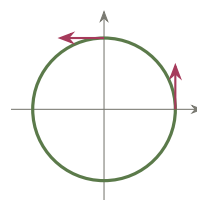
so the point circles anticlockwise while its radius decays exponentially with time constant 1. The speed is $\sqrt{2}$ times the radius — it slows in the same proportion as it closes in, which is why the spiral never arrives in finite time.



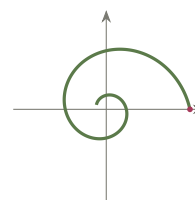
(a) line



(b) parabola



(c) circle



(d) spiral

The four paths, with velocity vectors in pink where they are worth seeing. (a) a straight line at constant speed $\sqrt{5}$; (b) the parabola $y = x^2$, speeding up as it steepens; (c) the unit circle at constant speed 1; (d) an inward spiral of radius e^{-t} and speed $\sqrt{2}e^{-t}$ — drawn with a slower decay than e^{-t} so that several turns stay visible.

Common mistake. Reporting $\|\mathbf{x}(t)\|$ when the question asks for $\|\dot{\mathbf{x}}(t)\|$ — in (d), answering e^{-t} (the radius) instead of $\sqrt{2}e^{-t}$ (the speed). Position and velocity are different vectors; the norm of one is a distance, the norm of the other is a rate. In (c) the two happen to coincide numerically, both being 1, which is exactly why (c) is a bad case to calibrate the habit on.

Exercise 8 $\mathbf{y}(t) = (\cos t, \sin t)$. (a) Show $\mathbf{y} \cdot \dot{\mathbf{y}} = 0$ for every t . (b) What does that say geometrically? (c) Velocity at $t = 0$ and $t = \pi/2$, marked on a sketch.

What this exercise uses. The dot product $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2$ is zero exactly when the two vectors are perpendicular. The identity used here is worth isolating, because it turns a statement about angles into a statement about lengths:

$$\frac{d}{dt} \|\mathbf{y}\|^2 = \frac{d}{dt} (\mathbf{y} \cdot \mathbf{y}) = 2 \mathbf{y} \cdot \dot{\mathbf{y}}.$$

So $\mathbf{y} \cdot \dot{\mathbf{y}} = 0$ and “ $\|\mathbf{y}\|$ is constant” are *the same statement*. Perpendicularity of velocity to position is not a coincidence of the circle; it is the definition of moving on one.

Solution.

(a) $\dot{\mathbf{y}} = (-\sin t, \cos t)$, so

$$\mathbf{y} \cdot \dot{\mathbf{y}} = \cos t (-\sin t) + \sin t (\cos t) = -\sin t \cos t + \sin t \cos t = \boxed{0}$$

for every t ✓.

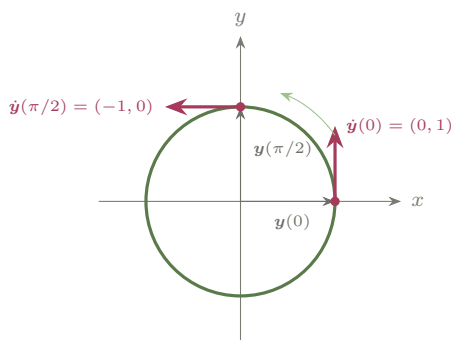
(b) The velocity is **always perpendicular to the position vector**. By the identity above, that is precisely the statement that $\|\mathbf{y}\|$ never changes: none of the motion is directed radially, so the distance to the origin cannot grow or shrink. All of the motion is tangential — which is what moving on a circle means.

(c) Substituting:

$$t = 0 : \quad \mathbf{y} = (1, 0), \quad \dot{\mathbf{y}} = (0, 1) \quad (\text{straight up}),$$

$$t = \frac{\pi}{2} : \quad \mathbf{y} = (0, 1), \quad \dot{\mathbf{y}} = (-1, 0) \quad (\text{straight left}).$$

In both cases the velocity is the position turned a quarter turn anticlockwise — consistent with (a), and with the anticlockwise sense found in Exercise 7(c).



At each instant the velocity (pink) is perpendicular to the position (grey), and tangent to the circle. The green arc marks the direction of travel.

Remark. Read the identity in the other direction and it becomes a tool: to show that some quantity stays bounded, show that its rate of change is orthogonal to it. That is the whole idea behind the Lyapunov function used in Exercise 15(c), where the same computation decides a case the eigenvalues cannot.

Exercise 9 Mouse-tracking: $\mathbf{x}(t) = (t, t^2(1-t))$ on $[0, 1]$, buttons at $(1, 0)$ and $(1, 1)$. (a) $\dot{\mathbf{x}}(t)$. (b) Velocity at $t = 0$ and $t = 1$. (c) Where is y largest, and what does that instant represent?

What this exercise uses. Two things from Session 3 applied to one component of a trajectory. First, $y(t)$ is maximised where $\dot{y} = 0$ and the sign of $\dot{y}$ changes from $+$ to $-$ — and on a *closed* interval the endpoints must be compared as well, since a maximum can sit at an end without the derivative vanishing there. Second, the velocity vector points along the direction of travel, so its components say where the cursor is heading at that instant. Expand before differentiating: $y = t^2(1 - t) = t^2 - t^3$.

Solution.

(a) Differentiating each component,

$$\dot{\mathbf{x}}(t) = \boxed{(1, 2t - 3t^2)}.$$

The horizontal component is constant: the cursor advances towards the buttons at a steady rate, and all the interesting behaviour is vertical.

(b) Substituting:

$$\dot{\mathbf{x}}(0) = \boxed{(1, 0)}, \quad \dot{\mathbf{x}}(1) = (1, 2 - 3) = \boxed{(1, -1)}.$$

At the start the cursor moves *straight at* the button $(1, 0)$, with no vertical component at all. At the end it arrives at $(1, 0)$ while still coming *down* at 45° — it is returning from a detour towards the other button.

(c) Set $\dot{y} = 0$ on $[0, 1]$:

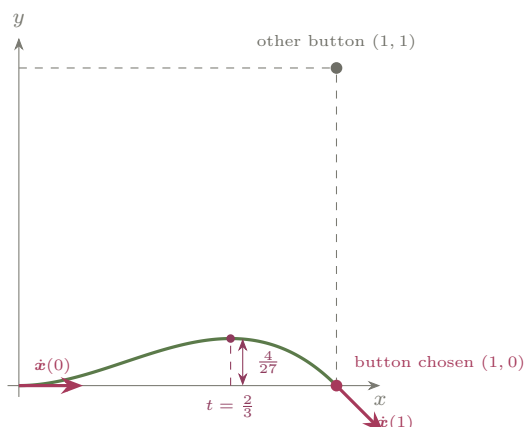
$$\dot{y} = 2t - 3t^2 = t(2 - 3t) = 0 \quad \implies \quad t = 0 \text{ or } t = \frac{2}{3}.$$

Compare the three candidates — the two endpoints and the interior critical point:

$$y(0) = 0, \quad y\left(\frac{2}{3}\right) = \frac{4}{9} \times \frac{1}{3} = \boxed{\frac{4}{27} \approx 0.148}, \quad y(1) = 0.$$

So the maximum is at $\boxed{t = 2/3}$. (Consistently, $\dot{y} > 0$ on $(0, 2/3)$ and $\dot{y} < 0$ on $(2/3, 1)$: up, then down.)

What that instant represents: the moment of **maximum deviation towards the unchosen button**. The cursor ends at $(1, 0)$, so $(1, 0)$ is the response given and $(1, 1)$ is the alternative; the bulge in y is the pull of that alternative. Its height, $4/27 \approx 0.15$, is about 15% of the way across — a visible but not decisive attraction. On this model the competition peaks two thirds of the way through the movement, and the last third is commitment.



The recorded path, drawn to scale. It leaves aimed exactly at the chosen button, bulges towards the unchosen one, peaks at $t = 2/3$ at a height of $4/27$, then comes back down and arrives still descending. The size of the bulge is the observable that a mouse-tracking study reports.

Common mistake. Answering (c) with $t = 0$ because it also solves $\dot{y} = 0$, or forgetting to check the endpoints at all. On a closed interval, “largest” means largest among the interior critical points *and* the two ends — three numbers here, and comparing them takes one line.

Exercise 10 Write $\mathbf{J}(x, y)$ and evaluate where asked: (a) $\mathbf{F} = (-x + y, 2x - 3y)$ (why is $\mathbf{J}$ constant?); (b) $\mathbf{F} = (x^2y, x + \sin y)$ at $(1, 0)$; (c) $\mathbf{F} = (x - x^3 - y, x - y)$ at $(0, 0)$ and $(1, 1)$; (d) for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, what is the Jacobian called?

What this exercise uses. The **Jacobian matrix** of $\mathbf{F} = (F_1, F_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ collects every partial derivative, one *output* per row and one *input* per column:

$$\mathbf{J}(x, y) = \begin{pmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{pmatrix}.$$

Row i is the gradient of the i -th component. The shape rule is worth stating once: for $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ the Jacobian is $m \times n$ — as many rows as outputs, as many columns as inputs — which is exactly what makes $\mathbf{J}\delta$ a legal product, taking an input perturbation to an output perturbation.

Solution.

(a) Each entry is a constant:

$$\mathbf{J} = \begin{pmatrix} -1 & 1 \\ 2 & -3 \end{pmatrix}.$$

It is constant because $\mathbf{F}$ is **linear**: $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$ with $\mathbf{A}$ this same matrix. The Jacobian is the best linear approximation, and the best linear approximation of something already linear is itself, at every point — so it cannot depend on where you evaluate it. (Compare one variable: $f(t) = at$ has $f' \equiv a$.)

(b) Differentiate row by row: $F_1 = x^2y$ gives $(2xy, x^2)$, and $F_2 = x + \sin y$ gives $(1, \cos y)$. Hence

$$\mathbf{J}(x, y) = \begin{pmatrix} 2xy & x^2 \\ 1 & \cos y \end{pmatrix}, \quad \mathbf{J}(1, 0) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

(At $(1, 0)$: $2xy = 0$, $x^2 = 1$, $\cos 0 = 1$.)

(c) $F_1 = x - x^3 - y$ gives $(1 - 3x^2, -1)$; $F_2 = x - y$ gives $(1, -1)$. So

$$\mathbf{J}(x, y) = \begin{pmatrix} 1 - 3x^2 & -1 \\ 1 & -1 \end{pmatrix},$$

$$\mathbf{J}(0, 0) = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}, \quad \mathbf{J}(1, 1) = \begin{pmatrix} -2 & -1 \\ 1 & -1 \end{pmatrix}.$$

Only the top-left entry moves, because x^3 is the only nonlinear term. These two matrices are used again in Exercises 11(c) and 15(c).

(d) With one output and two inputs the Jacobian is a 1×2 matrix — a single row, $(\partial_x f \ \partial_y f)$. That is the **gradient**, written as a row vector.

The distinction is bookkeeping, not mathematics: ∇f is usually written as a column so that $\nabla f \cdot \delta$ reads as a dot product, while the Jacobian convention makes it a row so that $\mathbf{J}\delta$ reads as a matrix–vector product. Both produce the same number.

Remark. The Jacobian is the single object that unifies this session: for one input and one output it is the number f' , for several inputs and one output it is the row ∇f , and in general it is the matrix above. Every statement about derivatives in Session 4 is a statement about it.

Common mistake. Transposing it — putting the partial derivatives of F_1 in the first *column* rather than the first *row*. In (a) the two conventions give different matrices ($\begin{pmatrix} -1 & 2 \\ 1 & -3 \end{pmatrix}$ instead of $\begin{pmatrix} -1 & 1 \\ 2 & -3 \end{pmatrix}$), and only one of them reproduces $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$. Check against the linear case whenever in doubt.

Exercise 11 (a) $\mathbf{F}(\mathbf{x}_0 + \delta) \approx ?$ (b) The one-variable statement it generalises. (c) Estimate $\mathbf{F}(0.1, -0.1)$ for $\mathbf{F} = (x - x^3 - y, x - y)$, given $\mathbf{F}(0, 0) = \mathbf{0}$.

What this exercise uses. Linearisation. Near a point, a differentiable map is well approximated by an affine one: the value at the point, plus a *linear* map applied to the displacement. That linear map is the derivative — in whatever shape the dimensions require. The error is of order $\|\delta\|^2$, so halving the displacement quarters the error, which is why the approximation is only claimed *near* $\mathbf{x}_0$.

Solution.

(a)

$$\mathbf{F}(\mathbf{x}_0 + \delta) \approx \mathbf{F}(\mathbf{x}_0) + \mathbf{J}(\mathbf{x}_0) \delta$$

— value at the base point, plus the Jacobian applied to the step.

(b)

$$f(t_0 + h) \approx f(t_0) + f'(t_0) h.$$

The two statements are the same sentence: “new value = old value + derivative $\times$ step”. Only the type of the derivative changes — a number becomes a matrix, and multiplication becomes matrix–vector multiplication.

(c) From Exercise 10(c), $\mathbf{J}(0, 0) = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$, and here $\delta = (0.1, -0.1)$ with $\mathbf{F}(0, 0) = \mathbf{0}$. So

$$\mathbf{J}(0, 0) \delta = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0.1 \\ -0.1 \end{pmatrix} = \begin{pmatrix} 1(0.1) + (-1)(-0.1) \\ 1(0.1) + (-1)(-0.1) \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.2 \end{pmatrix},$$

$$\text{hence } \mathbf{F}(0.1, -0.1) \approx \mathbf{0} + (0.2, 0.2) = \boxed{(0.2, 0.2)}.$$

Check against the exact value:

$$\mathbf{F}(0.1, -0.1) = (0.1 - 0.001 + 0.1, 0.1 + 0.1) = (0.199, 0.200).$$

The second component is exact — F_2 is linear, so the approximation is not an approximation at all. The first is off by 0.001, which is exactly the cubic term x^3 that the linearisation discards, and it is 0.5% of the answer.

Remark. Note what the error term tells you in advance: it is $-x^3$, so it is cubic in the step. Halving δ here divides the error by eight, not by four — this particular $\mathbf{F}$ has no quadratic term at all. That is why the estimate is so good, and it is a property of the function, not a general guarantee.

Exercise 12 For each matrix give $T = \text{tr } \mathbf{A}$, $D = \det \mathbf{A}$, the eigenvalues, and the flow type of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$: (a) $\begin{pmatrix} -2 & 0 \\ 0 & -5 \end{pmatrix}$; (b) $\begin{pmatrix} 1 & 0 \\ 0 & -3 \end{pmatrix}$; (c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$; (d) $\begin{pmatrix} -1 & -2 \\ 2 & -1 \end{pmatrix}$; (e) $\begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}$.

What this exercise uses. For a 2×2 matrix the characteristic polynomial is $\lambda^2 - T\lambda + D$ with $T = \text{tr } \mathbf{A} = a_{11} + a_{22}$ and $D = \det \mathbf{A} = a_{11}a_{22} - a_{12}a_{21}$, so the two numbers T and D contain the eigenvalues, and hence the whole classification. Reading the table on page 1: the sign of D decides saddle or not, the sign of the discriminant $T^2 - 4D$ decides node or spiral, and the sign of T decides stable or unstable. It is quicker to compute T and D than to solve for the eigenvalues, and for a classification they are enough.

Solution.

(a) $T = -2 - 5 = -7$, $D = (-2)(-5) = 10$. Diagonal, so the eigenvalues are read off: $\lambda = -2, -5$. Both real, distinct, negative: a **stable node**. Trajectories come in without turning, and along the direction with $\lambda = -5$ they arrive 2.5 times faster than along $\lambda = -2$. After a short while the second component — the fast one — is gone and the state is essentially $(x_1(0)e^{-2t}, 0)$: it settles onto the *first* axis and creeps in along that slow direction.

(b) $T = 1 - 3 = -2$, $D = (1)(-3) = -3 < 0$, $\lambda = 1, -3$: opposite signs, so a **saddle**, always unstable. Note that $T < 0$ here and the system is still unstable — $D < 0$ overrules everything.

(c) $T = 0$, $D = 0 \cdot 0 - (1)(-1) = 1$, so $\lambda^2 + 1 = 0$ and $\lambda = \pm i$: purely imaginary. A **centre** — closed orbits, neither growing nor decaying. (This is $\dot{x} = y$, $\dot{y} = -x$; from $(1, 0)$ the solution is $(\cos t, -\sin t)$, so it is the unit circle of Exercise 8 traversed *clockwise* — anticlockwise would be $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.)

(d) $T = -2$, $D = (-1)(-1) - (-2)(2) = 1 + 4 = 5$. Discriminant $T^2 - 4D = 4 - 20 = -16 < 0$, so

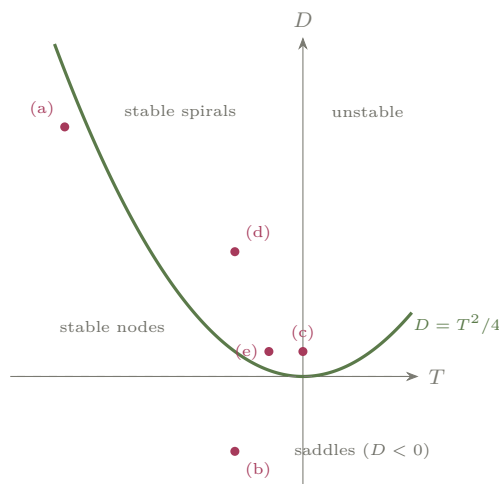
$$\lambda = \frac{-2 \pm \sqrt{-16}}{2} = -1 \pm 2i.$$

Negative real part and nonzero imaginary part: a **stable spiral**, rotating at angular speed 2 while decaying with time constant 1.

(e) $T = 0 + (-1) = -1$, $D = (0)(-1) - (1)(-1) = 1$. Discriminant $1 - 4 = -3 < 0$, so

$$\lambda = \frac{-1 \pm i\sqrt{3}}{2},$$

real part $-\frac{1}{2}$: a **stable spiral**.



The trace–determinant plane, with the five matrices placed on it. Above the parabola $D = T^2/4$ the eigenvalues are complex (spirals); below it and with $D > 0$ they are real of the same sign (nodes); below the axis $D = 0$ they are real of opposite signs (saddles). In the upper half $D > 0$ the sign of T decides: left of the D axis stable, right of it unstable, and the axis itself ($T = 0$, $D > 0$) carrying the centres. In the lower half everything is a saddle whatever the sign of T — which is case (b).

Remark. A **centre** is the one classification that does not survive linearisation. If the eigenvalues of a Jacobian are purely imaginary, the neglected nonlinear terms decide whether the true trajectories close up, spiral in, or spiral out, and the linear picture cannot tell you which. Every other case in the table — node, saddle, spiral, with nonzero real parts — is inherited by the nonlinear system near the fixed point. Exercise 15(c) is a second, even more degenerate instance of the same warning.

Common mistake. Concluding from $T < 0$ that the system is stable. Case (b) has $T = -2 < 0$ and is a saddle. T is the *sum* of the eigenvalues, and a sum can be negative with one term positive. Stability requires *both* $T < 0$ **and** $D > 0$ — the two conditions together, never T alone.

Exercise 13 (a) Verify that $\mathbf{x}(t) = e^{\lambda t} \mathbf{v}$ solves $\dot{\mathbf{x}} = \mathbf{A} \mathbf{x}$ whenever $\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$. (b) For $\mathbf{A} = \text{diag}(-1, -4)$ and $\mathbf{x}(0) = (2, 1)$, write $\mathbf{x}(t)$; which component dies first? (c) Eigenvalues $-0.1 \pm 3i$: stable? oscillating? at what angular speed and period? (d) Eigenvalues -100 and $+0.01$: stable?

What this exercise uses. An **eigenvector** $\mathbf{v}$ of $\mathbf{A}$ is a direction the matrix does not rotate: $\mathbf{A} \mathbf{v} = \lambda \mathbf{v}$, only stretched by the factor λ . Along such a direction the system $\dot{\mathbf{x}} = \mathbf{A} \mathbf{x}$ collapses to the scalar equation $\dot{s} = \lambda s$ of Session 3, whose solution is $e^{\lambda t}$ — so eigenvectors are the directions along which the two-dimensional problem is one-dimensional. A general solution is a *superposition* of such modes,

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2,$$

with c_1, c_2 fixed by the initial condition. For a complex pair $\lambda = \mu \pm i\omega$: the real part μ sets growth or decay (with $\tau = 1/|\mu|$), the imaginary part ω sets the rotation, at angular speed ω and period $2\pi/\omega$.

Solution.

(a) Differentiate the candidate ($\mathbf{v}$ is a constant vector, $e^{\lambda t}$ a scalar function):

$$\dot{\mathbf{x}}(t) = \lambda e^{\lambda t} \mathbf{v}.$$

Now apply $\mathbf{A}$ to the candidate and use the eigenvector property:

$$\mathbf{A}\mathbf{x}(t) = \mathbf{A}(e^{\lambda t}\mathbf{v}) = e^{\lambda t}\mathbf{A}\mathbf{v} = e^{\lambda t}\lambda\mathbf{v} = \lambda e^{\lambda t}\mathbf{v}.$$

The two expressions agree, so $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ ✓. (The middle step uses only that $\mathbf{A}$ is linear, so a scalar factor passes through it.)

- (b) The matrix is diagonal, so the two equations are already uncoupled: $\dot{x}_1 = -x_1$ and $\dot{x}_2 = -4x_2$. With $\mathbf{x}(0) = (2, 1)$,

$$\mathbf{x}(t) = (2e^{-t}, e^{-4t}).$$

The **second component dies first**: its time constant is $\tau_2 = 1/4 = 0.25$ against $\tau_1 = 1$, so it decays four times faster. Its smaller starting value (1 against 2) is irrelevant to the comparison — the rate is set by the eigenvalue alone, and after a few time units the state is essentially $(2e^{-t}, 0)$: the trajectory flattens onto the slow eigendirection before creeping in along it. *That is the general picture of a stable node: the fast mode dies, the slow mode dictates the approach.*

- (c) Real part $\operatorname{Re} \lambda = -0.1 < 0$, so **stable**. Imaginary part $\pm 3 \neq 0$, so it **oscillates**, at angular speed

$$\omega = 3 \text{ rad per unit time,} \quad \text{period} \quad \frac{2\pi}{\omega} = \frac{2\pi}{3} \approx 2.09.$$

The envelope decays with $\tau = 1/0.1 = 10$. So there are about $10/2.09 \approx 4.8$ oscillations per time constant: it *rings fast and settles slowly*, and a recording of it would look like a long, slowly fading wobble rather than a smooth return.

- (d) **Unstable**. Stability requires *every* eigenvalue to have negative real part; there is no averaging and no majority vote. The general solution is

$$\mathbf{x}(t) = c_1 e^{-100t} \mathbf{v}_1 + c_2 e^{+0.01t} \mathbf{v}_2,$$

and whatever c_1 is, the first term is gone almost immediately while the second grows without bound. The growth is slow — the time constant is 100, so it takes about 69 time units merely to double — which makes this the dangerous case in practice: a short simulation looks perfectly stable, because all one sees is the fast mode collapsing.

Common mistake. Averaging the eigenvalues, or looking at the trace: in (d) $T = -99.99$, which is very negative, and the system is unstable all the same. The trace is a sum, and it tells you about stability only in combination with the determinant — here $D = -1 < 0$, a saddle. When in doubt, look at the eigenvalues one at a time and ask of each: does this mode grow?

Exercise 14 Two mutually inhibiting accumulators: $\dot{x} = -x - cy$, $\dot{y} = -cx - y$, $\mathbf{A} = \begin{pmatrix} -1 & -c \\ -c & -1 \end{pmatrix}$, $c > 0$. (a) Eigenvectors without computing, then eigenvalues. (b) Interpret each eigenvector. (c) For which c does $x - y$ grow, and what does the model then do? (d) $c = 2$, $\mathbf{x}(0) = (1.1, 0.9)$: write $\mathbf{x}(t)$ and describe. (e) D for $c = 2$: confirm the flow type.

What this exercise uses. When a matrix has **equal diagonal entries and equal off-diagonal entries**, $\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$, its eigenvectors are always $(1, 1)$ and $(1, -1)$, whatever α and β : the matrix treats the two coordinates identically, so it cannot distinguish them, and the symmetric and antisymmetric combinations must be preserved. Multiplying out,

$$\begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (\alpha + \beta) \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = (\alpha - \beta) \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Once the eigen-directions are known, *decomposing the initial condition along them* turns the system into two independent scalar equations — which is the entire technique of this exercise.

Solution.

- (a) Here $\alpha = -1$ and $\beta = -c$, so the eigenvectors are

$$\mathbf{v}_1 = (1, 1) \quad \text{with} \quad \lambda_1 = \alpha + \beta = \boxed{-1 - c},$$

$$\mathbf{v}_2 = (1, -1) \quad \text{with} \quad \lambda_2 = \alpha - \beta = \boxed{-1 + c} = c - 1.$$

Check on the first: $\mathbf{A}(1, 1) = (-1 - c, -c - 1) = (-1 - c)(1, 1) \checkmark$. (They are orthogonal, as they must be: $\mathbf{A}$ is symmetric.)

- (b) $\mathbf{v}_1 = (1, 1)$ is the direction along which x and y move *together*: it measures the **total activity** $x + y$, how much evidence has accumulated overall. $\mathbf{v}_2 = (1, -1)$ is the direction along which they move *oppositely*: it measures the **difference** $x - y$, i.e. which of the two options is currently ahead.

The decomposition is therefore not a computational trick but the natural description of the model: a decision is a statement about the difference, not about the total.

- (c) The difference mode grows exactly when its eigenvalue is positive:

$$\lambda_2 = c - 1 > 0 \quad \Longleftrightarrow \quad \boxed{c > 1}.$$

Then any small initial imbalance is *amplified* — exponentially, at rate $c - 1$ — while the total activity keeps decaying at rate $1 + c$. The balanced state $x = y$ stops being an attractor and becomes a **saddle**: the model **commits** to whichever option was ahead, however slightly. For $c < 1$ both eigenvalues are negative, everything decays to $(0, 0)$ and the model never decides. So $c = 1$ is the threshold at which mutual inhibition turns an integrator into a decision-maker.

- (d) With $c = 2$: $\lambda_1 = -3$ along $(1, 1)$ and $\lambda_2 = +1$ along $(1, -1)$. Decompose the initial condition, $\mathbf{x}(0) = a(1, 1) + b(1, -1)$:

$$a + b = 1.1, \quad a - b = 0.9 \quad \Longrightarrow \quad a = 1.0, \quad b = 0.1.$$

Each mode then evolves on its own:

$$\boxed{\mathbf{x}(t) = 1.0 e^{-3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0.1 e^{+t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}} \quad \text{i.e.} \quad \begin{cases} x(t) = e^{-3t} + 0.1 e^t, \\ y(t) = e^{-3t} - 0.1 e^t. \end{cases}$$

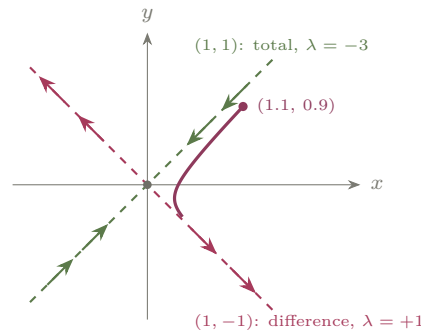
Check: $\mathbf{x}(0) = (1 + 0.1, 1 - 0.1) = (1.1, 0.9) \checkmark$.

Behaviour: the total activity $x + y = 2e^{-3t}$ collapses with time constant $\frac{1}{3}$, while the difference $x - y = 0.2e^t$ grows with time constant 1. Initially both accumulators fall (the decay dominates); then the difference takes over, x turns back up and y goes negative. The **x -accumulator wins** — the one that started 0.1 ahead — and it would have won from any initial condition with $x_0 > y_0$, however small the margin.

(e)

$$D = \det \mathbf{A} = (-1)(-1) - (-c)(-c) = 1 - c^2 \quad \implies \quad D|_{c=2} = 1 - 4 = \boxed{-3} < 0.$$

$D < 0$ means the eigenvalues have opposite signs: a **saddle** ✓, confirming (c) and (d). Note that $T = -2 < 0$ here, so the trace alone would have suggested stability — this is Exercise 12's warning in its modelling form. More generally $D = 1 - c^2 < 0$ exactly when $c > 1$, so the determinant reproduces the decision threshold of part (c) in one line.



The saddle at $c = 2$. Along $(1, 1)$ — total activity — the flow contracts fast; along $(1, -1)$ — the difference — it expands. The blue trajectory starts almost balanced, is pulled sharply towards the origin by the fast contracting mode, then swings out along the expanding direction: x up, y down. That swing is the decision.

Common mistake. Reading “both eigenvalues have negative diagonal entries” or “the diagonal is -1 , so everything decays”. The diagonal entries of $\mathbf{A}$ are not its eigenvalues unless the off-diagonal entries vanish. Here the off-diagonal coupling $-c$ is exactly what creates the instability, and it is invisible on the diagonal.

Remark. The threshold $c = 1$ is the model's substantive prediction: below it, accumulated evidence simply leaks away and no choice is made; above it, an arbitrarily small imbalance is amplified into a commitment. Which side a real system sits on is an empirical question, and it is decided by a single number — the ratio of inhibition to leak.

Exercise 15 (a) $\dot{x} = y, \dot{y} = -x - y$: fixed point, $\mathbf{J}, T, D$, classify. (b) $\dot{x} = y - x, \dot{y} = x^2 - y$: nullclines, fixed points, classify each. (c) $\dot{x} = x - x^3 - y, \dot{y} = x - y$: fixed point; show the linearisation is inconclusive. What next?

What this exercise uses. For a nonlinear system $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$:

- a **nullcline** is a curve on which one component of the velocity vanishes — the x -nullcline is $\{F_1 = 0\}$ (the flow crosses it vertically), the y -nullcline is $\{F_2 = 0\}$ (the flow crosses it horizontally);
- a **fixed point** is where *both* vanish, i.e. an intersection of the two nullclines. Drawing the nullclines is therefore how one *finds* the fixed points, not merely how one illustrates them;
- near a fixed point $\mathbf{x}^*$, writing $\mathbf{x} = \mathbf{x}^* + \delta$ and linearising gives $\dot{\delta} \approx \mathbf{J}(\mathbf{x}^*) \delta$. The classification of that linear system transfers to the nonlinear one **provided no eigenvalue has zero real part**. When one does, the linearisation is silent and the discarded terms decide.

Solution.

(a) $\dot{x} = 0$ gives $y = 0$; substituting into $\dot{y} = -x - y = 0$ gives $x = 0$. The only fixed point is

$(0,0)$. The system is already linear, so

$$\mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad T = 0 + (-1) = -1, \quad D = (0)(-1) - (1)(-1) = 1.$$

Discriminant $T^2 - 4D = 1 - 4 = -3 < 0$, and $T < 0$, so this is a **stable spiral**, with $\lambda = (-1 \pm i\sqrt{3})/2$. (It is matrix (e) of Exercise 12, and it is the equation $\ddot{x} + \dot{x} + x = 0$: a damped oscillator.)

(b) *Nullclines.* $\dot{x} = y - x = 0$ is the line $y = x$ (crossed vertically); $\dot{y} = x^2 - y = 0$ is the parabola $y = x^2$ (crossed horizontally). *Fixed points* are their intersections:

$$x = x^2 \iff x(x - 1) = 0 \implies (0,0) \text{ and } (1,1).$$

The Jacobian is

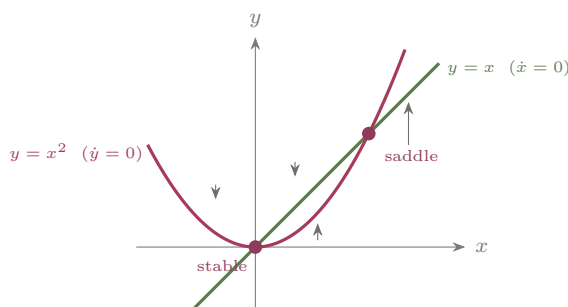
$$\mathbf{J}(x, y) = \begin{pmatrix} -1 & 1 \\ 2x & -1 \end{pmatrix}.$$

At $(0,0)$: $\mathbf{J} = \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$, so $T = -2$, $D = 1$, and $T^2 - 4D = 4 - 4 = 0$ — exactly on the borderline. The eigenvalue is $\lambda = -1$, repeated, and the matrix is *not* diagonalisable (it is a Jordan block: the only eigenvectors are multiples of $(1,0)$). This is a **degenerate (improper) stable node**: trajectories approach along the single eigendirection, with an extra factor t in the solution, $\delta(t) = e^{-t}(\mathbf{a} + t\mathbf{b})$. What matters for stability is that $\operatorname{Re} \lambda = -1 < 0$, so the point is **asymptotically stable** — and here the real part is nonzero, so the conclusion does transfer to the nonlinear system.

At $(1,1)$: $\mathbf{J} = \begin{pmatrix} -1 & 1 \\ 2 & -1 \end{pmatrix}$, so $T = -2$ and

$$D = (-1)(-1) - (1)(2) = 1 - 2 = -1 < 0 \implies \textbf{saddle},$$

with $\lambda = -1 \pm \sqrt{2}$, i.e. ≈ -2.41 and $\approx +0.41$: one direction in, one direction out. Unstable.



Nullclines of (b). They intersect twice, and those intersections are the fixed points; the grey arrows show the vertical component of the flow, which changes sign across the parabola. The origin attracts, $(1,1)$ repels along one direction.

(c) From $\dot{y} = x - y = 0$ we get $y = x$; substituting into $\dot{x} = x - x^3 - y$,

$$x - x^3 - x = -x^3 = 0 \implies x = 0,$$

so the *only* fixed point is $\boxed{(0,0)}$. From Exercise 10(c),

$$J(0,0) = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}, \quad T = 1 - 1 = 0, \quad D = (1)(-1) - (-1)(1) = -1 + 1 = 0.$$

With $T = D = 0$ the characteristic polynomial is $\lambda^2 = 0$: **both eigenvalues are zero**. Every eigenvalue has zero real part, so the hypothesis of the transfer principle fails as badly as it possibly can, and the linearisation says **nothing at all** — not “stable”, not “unstable”. (Its own flow is a pure shear along $(1,1)$, which is neither.)

What to do next. Two things, in this order.

1. Look at the nullclines and the flow. They are $y = x - x^3$ (vertical crossings) and $y = x$ (horizontal crossings); the two are tangent at the origin, which is the geometric face of the double zero eigenvalue.

2. Keep the cubic term, in the right coordinates. Set $u = x - y$, the combination along which the linearisation is degenerate. Then

$$\dot{u} = \dot{x} - \dot{y} = (x - x^3 - y) - (x - y) = -x^3, \quad \dot{x} = u - x^3.$$

Now build the quantity

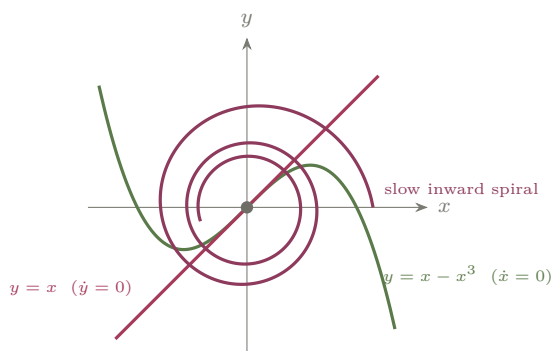
$$E = \frac{u^2}{2} + \frac{x^4}{4} \quad (\geq 0, \text{ zero only at the origin}),$$

and differentiate it along the flow — the same manoeuvre as Exercise 8:

$$\dot{E} = u\dot{u} + x^3\dot{x} = -ux^3 + x^3(u - x^3) = \boxed{-x^6} \leq 0.$$

So E can never increase: the origin is stable. And $\dot{E} = 0$ only when $x = 0$, where the equations force $\dot{x} = u$, so staying on $\{x = 0\}$ requires $u = 0$ too — the origin is the only trajectory on which E stops decreasing. Hence **the origin is asymptotically stable**: every nearby trajectory converges to it.

But not exponentially. Because $\dot{E} = -x^6$ is a very weak restoring term near 0, the decay is **algebraic**: numerically the amplitude behaves like $t^{-1/2}$. Since that makes the time to fall to a fraction f of the initial amplitude scale like f^{-2} , reaching a tenth takes about $(1/10)^{-2}/(1/2)^{-2} = 25$ times longer than reaching a half. No linear system does that — linear decay is always exponential — which is precisely why no linearisation could have produced this answer.



The two nullclines are *tangent* at the origin — that tangency is what the double zero eigenvalue looks like. The blue curve is a trajectory: it does converge, but its amplitude falls like $t^{-1/2}$ rather than exponentially, so it takes many turns to get anywhere. (The trajectory is schematic, drawn with an exaggerated decay so that the turns remain visible; the sense of rotation is the true one.)

Common mistake. Reading “the linearisation is inconclusive” as “the fixed point is unstable”, or as “ $T = 0$, so it is a centre”. Inconclusive means exactly what it says: the linear terms cancel out and the answer is carried by terms the linearisation threw away. In (c) the true answer is stability — the opposite of what a hasty reading of $T = 0, D = 0$ might suggest.

Remark. The three parts are a deliberate progression: in (a) the eigenvalues decide; in (b) they still decide, but one fixed point sits exactly on the node/spiral borderline, where the shape is delicate and the stability is not; in (c) they decide nothing and the nonlinear terms take over. Only the third case requires new tools, and that is the honest reason the linearisation theorem is always stated with its hypothesis attached.

Exercise 16 A single fixed point with $\mathbf{J} = \begin{pmatrix} -1 & \alpha \\ -1 & -1 \end{pmatrix}$. (a) T and D in terms of α . (b) For which α is it stable? (c) A spiral rather than a node? (d) A saddle?

What this exercise uses. Everything is read off the table on page 1, now with T and D depending on a parameter. The three questions are the three inequalities:

$$\text{stable} \iff T < 0 \text{ and } D > 0, \quad \text{spiral} \iff T^2 < 4D, \quad \text{saddle} \iff D < 0.$$

Because T here does not depend on α at all, the whole family lives on a single *vertical line* of the trace-determinant plane, and varying α slides the point up and down it.

Solution.

(a)

$$T = (-1) + (-1) = \boxed{-2}, \quad D = (-1)(-1) - (\alpha)(-1) = 1 + \alpha = \boxed{1 + \alpha}.$$

Explicitly, $\lambda_{\pm} = -1 \pm \sqrt{-\alpha}$.

(b) Stability needs $T < 0$ and $D > 0$. The first holds automatically ($T = -2$, whatever α), so the condition is $D = 1 + \alpha > 0$:

$$\boxed{\alpha > -1}.$$

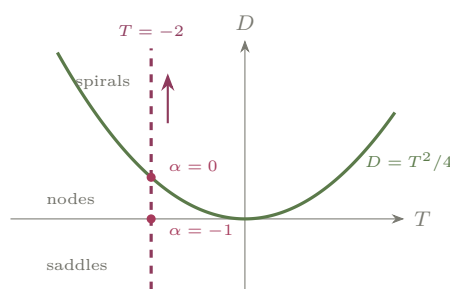
(c) A spiral needs complex eigenvalues, $T^2 < 4D$:

$$4 < 4(1 + \alpha) \iff 1 < 1 + \alpha \iff \boxed{\alpha > 0}.$$

(Consistently, $\lambda_{\pm} = -1 \pm \sqrt{-\alpha}$ is complex exactly when $\alpha > 0$, with imaginary part $\pm\sqrt{\alpha}$: the rotation speeds up as α grows.)

(d) A saddle needs $D < 0$:

$$1 + \alpha < 0 \iff \boxed{\alpha < -1}.$$



The whole family sits on the vertical line $T = -2$ (dashed). Since $T < 0$ throughout, the point is stable exactly on the part of the line *above* $D = 0$. Raising α raises $D = 1 + \alpha$ and slides the point upward: out of the saddles, across $D = 0$ at $\alpha = -1$, through the stable nodes, across the parabola at $\alpha = 0$, and into the stable spirals.

Summary, with the two borderlines included.

$\alpha < -1$	saddle ($\lambda_{\pm} = -1 \pm \sqrt{ \alpha }$, opposite signs)
$\alpha = -1$	$D = 0$: $\lambda = 0$ and -2 . The linearisation is degenerate — a whole line of fixed points for the linear system; inconclusive for a nonlinear one.
$-1 < \alpha < 0$	stable node (two real negative eigenvalues)
$\alpha = 0$	$T^2 = 4D$: $\lambda = -1$ twice. A degenerate (improper) stable node.
$\alpha > 0$	stable spiral ($\lambda = -1 \pm i\sqrt{\alpha}$)

The family is *never* unstable except as a saddle, because $T = -2$ is fixed: the sum of the eigenvalues is pinned at -2 , so they can never both be positive. What α controls is not whether the system settles but *how* — monotonically, or by ringing.

Common mistake. Answering (b) with “ $T < 0$, therefore always stable”. $T < 0$ alone rules out an unstable node and an unstable spiral, and says nothing about a saddle; for $\alpha < -1$ the point is a saddle and T is still -2 . Both inequalities, every time.

Remark. This is the shape of every parameter study in the session: fix the family, find where it sits in the (T, D) plane, and read off the values of the parameter at which it crosses one of the two boundaries $D = 0$ (a saddle appears) and $T = 0$ (stability is lost, and oscillations begin). Those crossings are what a modeller reports.